



OM 501 LA BlueTec[®] / OM 501 LA OM 502 LA BlueTec[®] / OM 502 LA

Operating Instructions



Mercedes-Benz

Symbols

 WARNING

Warning notes make you aware of dangers which could pose a threat to your health or life, or to the health and life of others.

 Environmental note

Environmental notes provide you with information on environmentally aware actions or disposal.

 Notes on material damage alert you to dangers that could lead to damage to your vehicle.

 These symbols indicate useful instructions or further information that could be helpful to you.

► This symbol designates an instruction you must follow.

► Several consecutive symbols indicate an instruction with several steps.

(▷ page) This symbol tells you where you can find further information on a topic.

▷ ▷ This symbol indicates a warning or an instruction that is continued on the next page.

Display This text indicates an indicator in the display.

Welcome

Familiarise yourself with your engine system and read the operating instructions before you use the engine system. This will help you to avoid endangering yourself or others.

The standard equipment and product description of your engine system may vary, depending on individual specifications. This is described on the data card.

The engine systems are constantly updated to be state of the art.

Mercedes-Benz reserves the right to make changes to the following:

- design
- equipment
- technical features

Descriptions may therefore differ in individual cases from your engine system.

Index	4	At a glance	13
Introduction	7	Safety	31
		Transport, installation	35
		Operation	39
		Maintenance	63
		Decommissioning	75
		Troubleshooting	79
		Technical data	93

A

AdBlue®/DEF	
Components	20
Consumption	57
Gauge	47
Important safety notes	59
Refuelling	59
Service product	71
ADM (FR (drive control) unit)	14

B

Battery (vehicle)	
Jump starting	83
Bio-diesel fuel	
see FAME fatty acid methyl ester fuel	
Bleeding the fuel prefilter with the manual fuel pump	82
BlueTec®	
Sensors	20
Braking	
Continuous brake	43
Engine brake	43
Retarder	44

C

Capacities	98
Care products	72
Charge current	44
Checking the fluid level	40
Cleaning and care	
Engine cleaning	73
High-pressure cleaning	72
Notes on care	72
Cold-start aid	60
Consumption	
AdBlue®/DEF	57
Fuel	57
Oil (engine)	57
Continuous brake	
Important safety notes	43
Coolant	
Mixing ratio	68
Service product	68
Topping up	41
Coolant additive	68

Correct use	8
Corrosion inhibitor/antifreeze agent	68

D

Data card	95
DEF/AdBlue®	
see AdBlue®/DEF service products	
Description of the engine	14
Diagnostics connection	33
Diesel	
Fuels	68
Low outside temperatures	70
Refuelling	58
Sulphur content table	69
Dimensions	95
Disposal of service products	65
Driving mode	
Idling speed	44
Driving tips	57

E

Electronic engine control	
see Engine management	
Electronic engine control unit	44
Emergency gearshift	
Using the emergency switch	46
Emergency running program	8
Engine	
Capacities	98
Cleaning	72
Data	95
Data card	95
Data plate	94
Modifying the power output	8
Oil consumption	57
Operating data	97
Rectifying faults	85
Running-in period	57
Starting	41
Stopping	42
Engine brake	43
Engine data	95
Engine data card	95
Engine description	14
Engine idling speed	44

Engine management	21
Engine management control unit	22
Engine model designation	95
Engine oil	
Consumption	57
For winter operation	65
Mixing	67
Oil change	66
Topping up	67
Engine overview	15
Engine speed	46
Exhaust gas aftertreatment	23
Exhaust gas aftertreatment control unit	23

F

FR (drive control) unit	
see ADM	
Fuel	
Additives	70
Consumption	57
Diesel	68
FAME fatty acid methyl ester fuel	70
Refuelling	58
Sulphur content table	69
Fuel grade	69
Fuel prefilter	
Draining	82
Fuel system	
Automatic bleeding	81
Manual bleeding	81
Fuses	
Checking and replacing a safety fuse	83
Important safety notes	82

G

Gauge	
AdBlue®/DEF	47
Genuine Mercedes-Benz parts	7
Getting started	80

H

High-pressure cleaning	72
-------------------------------------	----

I

Identification plate	94
Idling speed	
Engine	44
Indicator and warning lamps	
BlueTec® exhaust gas aftertreatment	26
Installation	36
Instrument cluster	
Warning and indicator lamps	25

J

Jump-starting	83
----------------------------	----

M

Maintenance	
Notes	64
Mercedes-Benz Service Centre	
see Qualified specialist workshop	
MR control unit	
see Engine management control unit	

O

Oil (engine)	
For winter operation	65
Oil change	66
Scope of use	66
Oil pressure	44
Operating data	97
Operating instructions	
General notes	10
Operating restriction	
Off-highway	28
On-highway	26
Operating safety	32
Operating safety and registration	
Changes in engine performance	8
Implied warranty	8
Operational monitoring	44
Organisational measures	33
Overview, sensors	19

P

Personnel	33
------------------------	----

Poly-V-belt	
Replacing	89
Routing	89

Preparing for starting operation

 see Starting operation

Protection of the environment	7
--	---

Q

Qualified specialist workshop	10
--	----

R

Refuelling

AdBlue®/DEF	59
Fuels	58

Requirements of the personnel	33
--	----

Rev counter	46
--------------------------	----

Roadside Assistance	80
----------------------------------	----

Running the vehicle in	57
-------------------------------------	----

S

Safety and emergency running pro- gram	8
---	---

Safety precautions	32
---------------------------------	----

SCR frame module

 see Exhaust gas aftertreatment
 control unit

Service products

AdBlue®/DEF	71
Coolant	68
DEF/AdBlue®	71
Diesel fuel	68
Disposal	65
Disposing of AdBlue®/DEF	72
Engine oil	65
FAME fatty acid methyl ester fuel	70
Fuel additives	70
General notes	65
Purity of AdBlue®/DEF	72
Storing AdBlue®/DEF	72

Specialist workshop	10
----------------------------------	----

Starting

 see Starting (engine)

Starting (engine)	41
--------------------------------	----

Stopping and switching off the engine	41
--	----

Sulphur content of fuel	69
--------------------------------------	----

T

Technical data

Dimensions	95
Filling capacities	98
Operating data	97
Weights	95

Tightening torques	99
---------------------------------	----

Torque reduction	45
-------------------------------	----

Transport	36
------------------------	----

W

Warning and indicator lamps

Electronics	48
-------------------	----

Weights	96
----------------------	----

Winter diesel	70
----------------------------	----

Winter operation	60
-------------------------------	----

Engine system

The OM 501-502 LA BlueTec® series of engines only function as intended when used in conjunction with the corresponding exhaust gas aftertreatment unit. Therefore, in these Operating Instructions, the term "engine system" refers to the engine and the exhaust gas aftertreatment unit.

Protection of the environment

Environmental note

Daimler AG has a declared policy of comprehensive environmental protection.

The objectives are to use the natural resources which form the basis of our existence on this planet sparingly and in a manner which takes the requirements of both nature and humanity into account.

You too can help to protect the environment by operating your vehicle in an environmentally responsible manner.

Information and notes on driving in an environmentally responsible and fuel-saving manner can be found in the "Operating notes" section (▷ page 57).

Assembly equipment

These Operating Instructions describe all models and all standard and optional equipment available for your engine system at the time of publication of the Operating Instructions. Country-specific deviations are possible. Note that your engine system may not be fitted with all features described. This also applies to safety-relevant systems and functions. Therefore, the equipment on your engine system may differ from certain descriptions and illustrations.

All of the components in your engine system are listed in the data card of your engine system. Data card (▷ page 95).

Please contact a Mercedes-Benz, MTU or MTU-authorised Mercedes-Benz Service Centre if you have any questions about equipment or operation (see addresses in the publication details on the rear inside cover).

- Mercedes-Benz recommends, for on-highway applications, that you use a Mercedes-Benz Service Centre.
- MTU recommends, for off-highway applications, an MTU partner or MTU-authorised Mercedes-Benz partner.

Genuine Mercedes-Benz parts

Environmental note

Daimler AG also supplies reconditioned assemblies and parts which are of the same quality as new parts. For these, the same warranty applies as for new parts.

If you use parts which have not been approved by Mercedes-Benz, the operational safety of the engine system may be jeopardised. This could lead to malfunctions in safety-relevant systems. Use only genuine Mercedes-Benz parts or parts of equal quality. Only use parts that have been approved for your engine type.

Mercedes-Benz checks genuine Mercedes-Benz parts for:

- reliability
- safety
- suitability

Despite ongoing market research, Mercedes-Benz is unable to assess other parts.

Mercedes-Benz therefore accepts no responsibility for the use of such parts in Mercedes-Benz vehicles, even if they have been officially approved or independently approved by a testing centre.

In Germany, certain parts are only officially approved for installation or modification if they comply with legal requirements. This also applies to some other countries. All genuine Mercedes-Benz parts meet the approval

requirements. The use of non-approved parts may invalidate the vehicle's general operating permit.

This is the case if:

- they result in a change to the vehicle type from that for which the vehicle's general operating permit was granted
- they pose a possible risk for road users
- they adversely affect the emission or noise levels

You can find more information on recommended conversion parts and accessories, as well as permitted technical modifications at any Mercedes-Benz, MTU or MTU-authorized Mercedes-Benz Service Centre (▷ page 10). Always state the engine number when ordering genuine Mercedes-Benz parts. You can find the engine number on the identification plate of your engine. You can also find the numbers on the data card (▷ page 95).

Modifying the engine output

! Increased power could:

- change emission levels
- cause malfunctions
- lead to consequential damage

The operating safety of the engine cannot be guaranteed in all situations.

Any tampering with the engine management system in order to increase the engine power output will lead to a loss of warranty entitlements.

Safety/emergency running program

The engine is equipped with an electronic engine management system that monitors the engine and has a self-diagnostic system.

If the electronic control system detects a malfunction, one of the following measures is automatically implemented after an appraisal of the malfunction:

- faults during operation are indicated by the corresponding warning lamp (▷ page 48).
- in conjunction with the electronic engine management system, fault codes with additional information can be shown on a display.
- the system switches to a suitable backup function for the continued, albeit restricted, operation of the engine. This includes torque and engine speed limitation, for example, as well as road speed limitation or constant emergency running speed.

Correct use

The engine system may only be installed as contractually specified.

The manufacturer of the end product is responsible for the correct installation of the engine and the exhaust gas aftertreatment system in the overall system.

The engine and the exhaust gas aftertreatment system may not be modified. If the engine is modified, Mercedes-Benz and MTU do not accept responsibility for any damage arising as a result.

Correct use of the engine system also requires adherence to the instructions in these Operating Instructions. This also requires compliance with the maintenance intervals and the professional execution of maintenance work. Please observe the Workshop Information System (WIS) (▷ page 10).

Implied warranty

A well-developed network of Mercedes-Benz, MTU and MTU-authorized Mercedes-Benz Service Centres is available to carry out maintenance work.